

Lecture -3 Data Structure/Distributed Systems Blockchain and Cryptocurrency

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Merkle Tree

- Blocks uses the Merkle Tree to store the transaction details, account balances, etc.

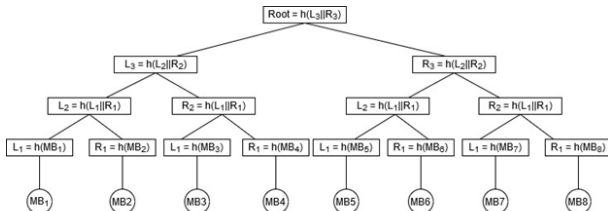


Figure: Merkle tree

Bloom Filter

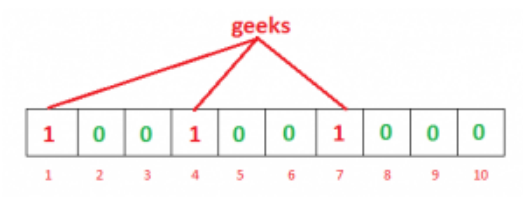


Figure: Bloom Filter [geeksforgeeks]

Distributed System

- A distributed system is a system whose components are located on different networked computers, which communicate and coordinate their actions by passing messages to one another.
- The components interact with one another in order to achieve a common goal.

Problem

- How to achieve common goal? Consensus

Byzantine Generals Problem

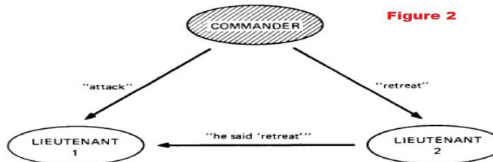


Figure: The problem was explained aptly in a paper by LESLIE LAMPORT, ROBERT SHOSTAK, and MARSHALL PEASE at Microsoft Research in 1982

Problem

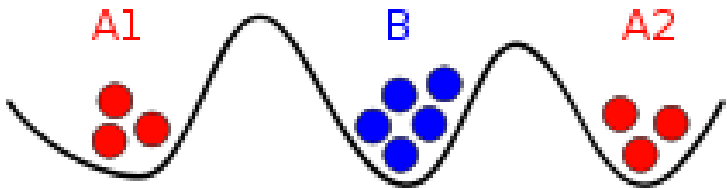


Figure: TGP

Network Types based on communication

- Synchronous Network: With same time.
- Asynchronous Network: Variation in the time.

Byzantine Fault Tolerance (Practical)

- Based on Majority.

$$N \geq 3t \quad (1)$$

Problem

- It may not work when more than t are faulty?

Miguel Castro and Barbara Liskov, 1999

Consensus Protocol

- Probabilistic
- Accurate

Proof of Work (PoW)

- This is to control the rate at which blocks can be generated
- This is added in the bitcoin to make the miners to do some work before constructing the blocks.
- Due to the very low probability of successful generation, this makes it unpredictable which worker in the network will be able to generate the next block.

$$n \leq 2^{256}/H_d$$

Proof of Stake (PoS) — Proof of Burn (PoB) [For other coin] — Proof of Elapsed Time (PoE)

- To reduce the energy consumption.

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Thank You!